GAYATHRE V

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Summary

Data Science graduate student with expertise in machine learning, deep learning, LLM systems, and RAG pipeline development. Delivered 14+ end-to-end AI/ML projects spanning quantitative finance, contract intelligence, anomaly detection, and medical imaging each deployed publicly with live demos. Proficient across the full ML lifecycle: model training, explainability (SHAP, Grad-CAM), frontend delivery (React, Next.js), and MLOps (MLflow, Docker, Redis). Published researcher with an accepted conference paper on NLP and fairness.

Education

Integrated M.Sc. in Data Science — Amrita Vishwa Vidyapeetham, Coimbatore
Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Statistical Modelling, Data Mining, Computer Vision — CGPA: 7.77 / 10.00

Higher Secondary Education (CBSE) — Jawahar Higher Secondary School — Marks: 402 / 500

Research & Publications

Bias-Aware Toxicity Detection in Social Networks — ***Accepted***
International Conference on Signal, Systems and Computing for Next-Gen Automation (ICSSCA) — *Pending Presentation*
Python, NLP, Deep Learning, Scikit-learn, Transformer Models, Fairness AI, Jigsaw Dataset

- Developed a bias-aware NLP pipeline using the Jigsaw Unintended Bias in Toxicity Classification dataset, applying a tri-hybrid modelling approach that combines traditional ML, deep learning, and transformer-based models.
- Evaluated model performance through bias-sensitive fairness metrics, demonstrating improved equitable detection across demographic identity groups including race, gender, and religion.
- Designed to reduce false positive rates on identity-mentioning text — a key challenge in content moderation systems.

Alzheimer's Disease Classification via Hybrid Deep Learning Ensembles — *Under Review*
Python, TensorFlow, Keras, PyTorch, Hugging Face, OpenCV, Grad-CAM, Vision Transformer (ViT)

- Achieved 99.81% classification accuracy on the OASIS MRI dataset using a Dual VGGNet ensemble, benchmarked against 8 architectures including ResNet50, EfficientNet, and VGG19+ViT across 4 heterogeneous datasets.
- Validated clinical relevance via Grad-CAM and Saliency Maps, verifying model attention on hippocampal atrophy and ventricular enlargement — primary neurological markers of Alzheimer's progression.

Projects

LegalLens — AI Contract Intelligence System (RAG Pipeline) — Live
Python, LLaMA 3.3 70B, Groq API, ChromaDB, Sentence Transformers, LangChain, Streamlit

- Engineered a 4-agent RAG pipeline (Clause Extractor, Risk Scorer, Obligation Tracker, Summariser) that reduces contract review time from hours to under 2 minutes across NDA, employment, and SaaS contracts.
- Achieved zero inference cost by routing all LLM calls through Groq's free-tier LLaMA 3.3 70B; combined LangChain recursive chunking, all-MiniLM-L6-v2 embeddings, and ChromaDB vector search.
- Deployed on Streamlit Cloud with live demo; accurately surfaces high-risk clauses including auto-renewal traps and uncapped liability terms.

RepoMind — AI-Powered Code Intelligence (RAG + LangGraph) — Live
Python, RAG, LLaMA 3.3 70B, ChromaDB, Sentence Transformers, LangGraph, GitPython, Streamlit

- Built a RAG-based tool that ingests any public GitHub repository and enables natural language Q&A, bug detection, security vulnerability scanning, AST-based dependency mapping, and auto-documentation generation.
- Processed 100+ file repositories via semantic chunking and embedding pipeline (all-MiniLM-L6-v2 + ChromaDB) with LangGraph orchestration and LLaMA 3.3 70B for zero-cost inference.

Optix — Options Strategy Payoff Visualizer & Risk Engine

Next.js 14, FastAPI, D3.js, Zustand, WebSockets, Redis, Black-Scholes, Docker Compose, Tailwind CSS

- Developed a full-stack quantitative finance platform supporting real-time construction and risk analysis of 6+ multi-leg options strategies with live P&L surfaces, breakeven points, and Greeks dashboards (delta, gamma, theta, vega).
- Engineered sub-100ms Greeks updates via WebSocket streaming with Redis caching; containerised full stack with Docker Compose for single-command deployment.
- Visualised theta decay using animated D3.js payoff curves, making time-value erosion intuitive for non-specialist users.

StockSense — AI-Powered Stock Anomaly Detection

Python, FastAPI, React, PyTorch, Scikit-learn, SHAP, MLflow, yfinance, lightweight-charts

- Built a dual-model anomaly detection system combining Isolation Forest and LSTM Autoencoder to flag statistically unusual trading days; SHAP analysis revealed Z-score and volatility drove 70%+ of detections.
- Tracked all experiment runs, parameters, and metrics via MLflow; served results through a FastAPI backend with a React frontend featuring TradingView-style OHLC candlestick charts and per-model anomaly markers.

Telco Customer Churn Prediction Dashboard Live

Python, XGBoost, LightGBM, Random Forest, Scikit-learn, SHAP, SMOTE, Streamlit

- Achieved 84%+ ROC-AUC on telecom churn prediction by ensembling XGBoost, LightGBM, and Random Forest on SMOTE-balanced data with 12 engineered features, improving minority class recall by approximately 18%.
- Deployed an interactive Streamlit dashboard featuring real-time SHAP explainability, counterfactual what-if analysis, and an industry-standard model card for non-technical stakeholders.

Technical Skills

Programming Languages: Python, R, SQL (MySQL), MATLAB

Machine Learning: Scikit-learn, XGBoost, LightGBM, Random Forest, SHAP, SMOTE, NumPy, Pandas, SciPy, Isolation Forest, AutoML

Deep Learning & NLP: PyTorch, TensorFlow, Keras, Hugging Face Transformers, LSTM, Vision Transformer (ViT), XLM-RoBERTa, mBERT, ELECTRA, Transfer Learning

LLM & RAG: LLaMA 3.3 70B, Groq API, Anthropic Claude, ChromaDB, Sentence Transformers, LangChain, LangGraph, Prompt Engineering, Multi-Agent AI, Vector Databases

Web & API Development: FastAPI, Flask, React, Next.js 14, Streamlit, WebSockets, REST APIs, Server-Sent Events (SSE)

MLOps & Cloud Infrastructure: MLflow, Docker, Docker Compose, Redis, Git, Azure App Service, Azure Blob Storage, Azure SQL

Data Visualisation & Explainability: Matplotlib, Seaborn, Plotly, D3.js, OpenCV, Grad-CAM, Saliency Maps, SHAP, LIME

Certifications

Google Data Analytics Professional Certificate — Coursera (2024)
Hands-on training in data cleaning, exploratory data analysis (EDA), SQL querying, statistical analysis, and data visualisation using real-world datasets across 8 courses.

Languages

English — Full Professional Proficiency **Hindi —** Full Professional Proficiency **Tamil —** Limited Working Proficiency